ONR YIP technical proposal: panel review

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Scope and conventions

This reviews printed pages 1–20 (Sections 1–6) of main-full.pdf at commit 029dff1 (Oct 28, 02:11), whose PDF and .tex match. Equations were read from the .tex.

- **Revised Oct 28 after an independent check.** The first version's space estimates were about 20 lines short, and a few fixes were wrong as written. Every fix below is corrected, applied to a copy of main-full.tex, and compiled: Sections 1–6 end on p. 20 with about 4 lines to spare (Space plan, at the end). The compiled source is main-full-reviewed.tex.
- Page numbers are the printed ones; printed p. 1 is the third PDF page.
- Comments run C1–C63 in reading order. C4–C13 are the ranked problems; C49–C63 are line edits. Section numbers follow your request, which has no item 5.
- The white paper used as the "before" is white_paper_predrag-4page-v4.pdf from the repo. The attached white_paper_predrag.pdf is the 1-page Oct 13 note.
- Panel shorthand: M = the mathematician or statistician, A = the AI researcher, N = the Navy S&T reviewer.

1. Panel verdict

Borderline on the technical proposal alone. It becomes fundable if the novelty credits and the math framing are fixed and two pilots are run; as submitted, the statistician on the panel pulls it down.

Technical merit: Very Good (M alone would score it Good). Panel summary: "A coherent program on a timely problem, aggregating judgments of LLM agents whose errors are correlated and some of whom may be adversarial, organized around one question and a running maritime example. The mathematics is thinner than claimed: four of six propositions follow in a line from the definitions, the central rater model is that of Jaffe et al. (2016) and Proposition 6 is a known noisy-reward identity, neither credited, and the nontrivial guarantees are deferred. Direct evidence for the proposed methods is one simulation; the other results are placeholders or prior work on adjacent methods."

Naval relevance: Very Good. Panel summary: "The watch-officer example is concrete, and fusing reports of unknown, correlated, possibly forged provenance maps credibly to maritime domain awareness, intelligence analysis, and AI test and evaluation, in line with AI and Autonomy and the 2026 ONR S&T Strategy's call for independent AI critique. All testbeds are public or synthetic, no Navy data or partner is named, and transition rests on a statement that NRL could evaluate the methods."

The one change that most raises the odds: make the theory honest and load-bearing in one pass. Credit Jaffe et al. (2016), Cai et al. (2025) and Chow (1970); call Propositions 1–6 design lemmas; and state four open questions as the research at the end of Section 3.1 (C4, C5). This answers the program officer's "ad hoc" comment and the NOFO's "awareness of the state of the art" criterion together, and fits the page limit with the cuts named in C4 and C5 (compile-tested).

Strengths not to edit away

- **C1** (pp. 1–2, Figure 1). One central question and one running Navy example threaded through every task and the figure. This is the structure the program officer praised. Keep the example's specifics (06:00 schedule, 09:10 port report, berth 4).
- **C2** (pp. 1, 6). The chain from failures F1–F4 to tasks to the terms of eq. (1), with rules R1–R5. R2 (ratings pick who argues; the judge picks the answer), R4 (nothing triggers a human audit) and R5 (planted cases never enter training) are crisp, testable, and the best structural answer to "ad hoc".
- **C3** (pp. 9, 16–18, Table 4). The evaluation includes regimes where the propositions predict failure: fewer than three independent groups, a colluder that cheats once, no tool. With red-team levels 1–4 and safety–usefulness curves, this is what "skeptical and falsifiable" looks like for Gold Standard Science.

2. Top problems

Four must-fix problems (C4–C7) decide the outcome. All ten fixes, worded as below, fit pages 1–20 together with the cuts in the Space plan; I tested this by compiling a copy of main-full.tex.

#	Problem	Priority	In the tested plan
C4	Rater model and Proposition 6 are known results, presented as new	Must-fix	Yes

#	Problem	Priority	In the tested plan
C5	Propositions are design lemmas; the real theory is deferred	Must-fix	Yes
C6	RG3 guarantees fail as stated; wrong decisions can reinforce persuasive messages	Must-fix	Yes
C7	Five pilot results are unmeasured; two would not be evidence	Must-fix	Yes; measured results in your original sentences leave 2 spare lines
C8	Colluders can win without a debate; eq. (1) hides it	Should-fix	Yes
C9	Task 1.2′s price bound falls with the number of groups, against the wrong benchmark	Should-fix	Yes
C10	Judge trust answered in pieces; Figure 1 never shows the real threat	Should-fix	Yes
C11	Recent work a reviewer will name	Should-fix	Yes
C12	Overlap boundary is vague	Should-fix	Yes; you supply the grant′s scope
C13	Success criteria and Navy payoff have no numbers	Optional	Uses 2 of the 4 spare lines

C4. The rater model and Proposition 6 are known results, presented as new · must-fix

Where: §1.3 (i), p. 3; §2, p. 4; Table 1 caption, p. 6; Task 1.1, p. 7; eq. (3), p. 9; Proposition 6, p. 14.

Reviewer (M): "Eq. (2) is the dependent-classifier model of Jaffe et al. (2016): classifiers split into groups, each group has a latent variable that depends on the true label, and members are independent given it. The proposal cites that paper only for 'dependence between classifiers', then proposes the same model as new. Proposition 6 is the symmetric case of the noisy-reward identity in Cai et al. (2025), itself the classical noisy-label correction of Natarajan et al. (2013). Rule (3) is Chow's (1970) reject rule."

Fix:

- Task 1.1, p. 7. Replace "We propose a shared-factor rater model in which a group that shares a mistake counts at most once." with: "We extend the dependent-classifier model of Jaffe et al. (2016) to more answers and to agents that argue and may be adversarial; in it, a group that shares a mistake counts at most once."
- p. 9, after "…so higher stakes lead to more debates.", add: "Rule (3) is the reject rule of Chow (1970), with debate as an imperfect expert (Mozannar and Sontag, 2020)."
- p. 14, Proposition 6's title: "(decisions as noisy labels)" → "(decisions as noisy labels; cf. Cai et al., 2025)", written \citealp[cf.][]{cai2025noisy}. This credits the result in a title, at no cost in lines.
- pp. 3–6: C55 and C60 credit Jaffe et al. in §1.3, §2 and Table 1; C62 drops a criticism of Yang et al. that would rebound on reward (5).
- The new bibliography entries are listed in the Space plan and already in main-full-reviewed.tex; they cost no page space.

What stays new, said plainly: a posterior that decides which questions to debate, unanimous ones included; audit flags inside the rater model; wrong answers that concentrate rather than spread; and the partition questions (Q1 in C5).

Cut: p. 9, delete "Our Peer Rank, which weights each AI reviewer by how highly the others rate its answers, agreed with human preferences more often than the strongest single judge (67.3% vs. 64.3% for GPT-4) (Li et al., 2024a); track records keep models that share mistakes from raising one another's standing." PRD did not show the last clause, and Task 1.2 already uses PRD.

C5. Most propositions are design lemmas, and the real theory is deferred · must-fix

Where: §1.3 "Why it will succeed", p. 3; §3.1, pp. 6–7; Propositions 1, 3, 4, 5 (pp. 8, 10, 12, 13).

Reviewer (M): "Propositions 1, 3, 4 and 5 follow in a line or two from the definitions, and Proposition 2 is a corollary of Allman et al. (2009). They formalize the design, which helps, but they are not the guarantees this program funds. The hard questions appear only as things 'we will characterize', so 'a principle whose property we prove' overstates what is shown."

Fix. At the end of §3.1, after "and our ablations measure each term (Section 3.5).", add:

"**Open questions.** Propositions 1–6 are design lemmas; the project's theory is four results we do not yet have: (Q1) how many questions it takes to recover a shared-error group, as a function of its size, λ_g, and how rarely others give its wrong answers, and when the

partition is identifiable (Task 1.1); (Q2) the largest gain of a coalition of f raters spread across groups (Task 1.2); (Q3) how the chance that the right side finds a false step falls with argument length, which decides when recursion keeps ε_step small (Task 2.1); and (Q4) the excess wrong decisions a colluder causes before it is flagged, expected to be of order √(n log log n / ρ) after n decisions at audit rate ρ (Task 2.2)."

Then fix §1.3 (C57). The labels can stay "Proposition"; the sentence above does the reframing.

Cut:

- p. 12. Delete "In our peer discussion, agreement with human preferences rose from 72.9% (GPT-4 alone) to 75.0% when GPT-3.5 led the discussion but fell to 71.4% when GPT-4 led (Li et al., 2024a): one reviewer can carry a discussion." It motivates F2 and says nothing about the harness.
- p. 9. Replace "Because Proposition 2 fixes the partition, we will characterize how many questions it takes to detect a group, as a function of its size, λg, and how rarely others give its wrong answers, and when the partition itself is identifiable." with "Proposition 2 fixes the partition; recovering it is Q1."
- p. 8. Shorten the paragraph after Proposition 1 to "So a group carries at most the weight of one agent that errs with probability λ_g; with |Y| answers and wrong answers equally likely, the bound is log[(|Y|−1)(1−λ_g)/λ_g]."
- p. 13. Fold Proposition 5′s proof into its statement (C19).

C6. RG3's guarantees fail as stated, and wrong decisions can still reinforce persuasive messages · must-fix

Where: §3.4, pp. 14–16; eqs. (5)–(6); Proposition 6; the Task 3 pilot.

Reviewer (A): "Four problems. Under policy gradient the chance term in (5) does not depend on the action, so it is a baseline and cannot stop colluders who bias the reference. Under GRPO′s per-question normalization, scaling a question′s rewards by ω_q does nothing. The weight 1 − η|Y|/(|Y|−1) assumes wrong answers are equally likely, the opposite of the shared-mistake model. And credit (6) rewards moving the judge toward a decision those same messages produced, the very reinforcement the colleague asked about."

Counterexamples a reviewer can write down:

- **Chance term.** Binary questions, half of them yes. Colluders outside agent a's group always say yes and dominate the reference. Truthful answers and "always yes" both earn zero expected reward under (5), but on every no-question the first term pushes a toward yes and the second adds no gradient. Training drifts to "always yes".
- **Multi-answer weight.** Four answers; the reference gives the truth with probability 0.45 and one shared wrong answer with 0.55. The formula gives $\omega = 1 - 0.55 \times 4/3 \approx 0.27 > 0$, yet agreement pays more for the wrong answer.
- **One multi-answer step.** The reference gives (y, w_1, w_2) with probabilities (0.55, 0.40, 0.05), and the policy puts (0.01, 0.98, 0.01) on them. Answering y maximizes the expected reward, yet the expected REINFORCE step lowers the probability of y. "Does not reverse" holds only for yes-or-no questions.

Fix, p. 15. Replace the passage from "So a wrong decision shrinks the update" through "even when colluders coordinate on them." with:

"So a wrong decision shrinks the update but does not reverse it while $\eta_q < 1/2$; with more answers, truthful answering maximizes the expected reward when the reference's most likely answer is the true one. The weight ω_q, the left-out posterior's margin between its top two answers ($1 - 2\hat{\eta}_q$ for yes-or-no questions), counts confident references fully and near-ties little; under GRPO it scales the loss, since per-question normalization would cancel it in the reward. Independence fails when the reference includes the agent's own answer, which is how rewards for majority agreement collapse (Shafayat et al., 2025); leaving out the agent's group restores it under the model of Task 1.1, and leaving out flagged agents guards it against colluders, which the chance term alone does not."

In symbols, the weight is below. It equals $1 - 2\hat{\eta}_q$ for yes-or-no questions and is nonnegative by construction, so it weights confidence but cannot detect a reference whose top answer is wrong; only the held-out check on known answers can (p. 16).

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\omega_q=\hat P^{-a}\big(y_q=D_q^{-a}\big)-\max_{w\neq D_q^{-a}}\hat P^{-
a}\big(y_q=w\big)
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Fix, p. 15, in the paragraph on (6). Replace "When the judge decides by re-running evidence, the decision does not depend on persuasion, so messages earn credit for moving the judge toward what the evidence shows." with:

"Message credit inherits Proposition 6 only when the decision's error does not depend on the messages, roughly when the judge settled the dispute by re-running a tool whose error does not depend on which evidence was chosen; we therefore credit messages only in such debates and, after debates settled by argument alone, train only first answers."

In the same paragraph, change "and $\hat{\eta}_q$ here comes from that probability" to "and ω_q here is that probability's margin". The new rule restores the white paper's "tool-backed decisions weigh more", now with a reason, and matches Figure 1, which already credits A4's tool-backed step. In the pilot, delete "close to the $1 - 2\hat{\eta} = 0.74$ that Proposition 6 predicts" (C20, C32). Optional: C53 and C56 align §1.2 and §1.3 at no cost in space.

Cut:

- p. 15. Shorten "At scale, we will train a multi-agent process reward model … such as Shapley values." to "At scale, a multi-agent process reward model trained on logged debates, extending our FG-PRM, will predict each message's credit."
- p. 15. Shorten the FG-PRM and HiPRAG sentence to "Our FG-PRM detects six types of step-level hallucination (Li et al., 2025), and HiPRAG rewards each search decision (Wu et al., 2026)."
- p. 19. Shorten "Reporting and dissemination" to "The PI will submit annual and final reports, attend Program Reviews, and release code at M1, M6, and with each paper."
- p. 15. The pilot reworded as an expected outcome (C32) is also shorter.

C7. Five results are unmeasured, and two would not be evidence if measured · must-fix

Where: orange text on pp. 9, 10, 12, 14, 15.

Reviewer (all): "If these numbers are results, where do they come from? Without them, the direct evidence is one simulation. Two would not count even when measured: 'not at all under our mechanism' (p. 10) is Proposition 3(a), true by construction, and Proposition 6 predicts no '0.74 of the gain' (p. 15)."

Fix: run the Task 2.1 and Task 1.1 pilots (C33), and restate the other three as expected outcomes with what would refute them (C29, C31, C32). Never delete only the \pending wrapper and keep the numbers.

Cut: none; the expected-outcome wording is shorter.

C8. Colluders can win without a debate, and eq. (1) hides it · should-fix (close to must)

Where: eq. (1), p. 6; text, p. 7; Proposition 2, p. 8; the colleague's question (a).

Reviewer (M): "The first term of (1) assumes a calibrated posterior, and dishonest agents enter only the debate branch, as ε_{role}. Colluders spread across groups can push a wrong answer above τ so it is never debated, and under Proposition 2's independence they look

like independent confirmation. Nothing in (1) bounds this."

Fix. Keep (1) as it is and add one term after the first, labeled "Tasks 1.1, 2.2". It still sets on one line, with no overfull box:

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P(D_q\neq y_q)\;\le\;\underbrace{(1-\tau)\,P(\pi_q\ge\tau)}_{\text{no debate: Task 1.1}}\;+\;\underbrace{\epsilon_{\text{cal}}}_{\text{Tasks 1.1, 2.2}}\;+\;P(\pi_q<\tau)\,\big[\underbrace{\epsilon_{\text{cand}}}_{\text{Task 1.1}}+\underbrace{\epsilon_{\text{role}}}_{\text{Tasks 1.2, 2.2}}+\underbrace{\epsilon_{\text{step}}}_{\text{Task 2.1}}\big]
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- p. 7. Replace "The first term holds when the posterior is calibrated (answers given probability π are right a fraction π of the time); a vote that ignores shared mistakes is overconfident and breaks it (Figure 2(a)), so Task 1.1 supplies a posterior that accounts for them and sets τ by weighing a debate's cost against the stakes." with: "Here $\varepsilon_{cal} = E[(P(\hat{y}_q \neq y_q \mid \pi_q) - 1 + \pi_q)_+ 1\{\pi_q \geqslant \tau\}]$ is the error beyond what the posterior claims on questions that skip debate, so (1) needs no calibration assumption. It comes from shared mistakes the model misses, which make a vote overconfident (Figure 2(a)), or from colluders who push a wrong answer past τ; random human audits estimate it, and Tasks 1.1 and 2.2 shrink its two sources. Task 1.1 also sets τ by weighing a debate's cost against the stakes."
- p. 8, after Proposition 2's last sentence ("…cannot tell an accurate majority from one that copies a mistake."), add: "Colluders who coordinate across groups also break this independence; audits limit the harm they do before they are flagged (Task 2.2), and a flagged coalition is merged into one group."
- Optional, after C4's Chow sentence: "For a calibrated posterior and ε_D that does not vary with π_q, (3) minimizes $L \cdot P(D_q \neq y_q) + c \cdot P(\pi_q < \tau)$, with ε_D at most the bracket in (1)." It turns (1) into an objective and costs 2 of the 4 spare lines.

Cut:

- Table 1, p. 6. Merge the three all-✗ rows (Single AI judge; Majority vote, juries; Open discussion among LLMs) into one: "Single judge, vote, juries, open discussion (Zheng et al., 2023; Verga et al., 2024; Du et al., 2024)".
- §3.7, p. 19. Shorten "Without conflicts of interest" to "The PI discloses outside activities and support under UT Dallas policy."

C9. Task 1.2's price bound falls with the number of groups, against the wrong benchmark · should-fix

Where: Task 1.2, pp. 9–10: eq. (4), Proposition 3(b), Figure 3; Figure 1 panel 2.

Reviewer (M, social choice): "Proposition 3(b) guarantees 1/|E_y| of the best arguer's quality, which falls as more groups give an answer, and inside a one-member group ratings play no role, so when groups are small the debater is close to a random draw. The comparison with Kurokawa et al.'s 1/(1+m) mixes benchmarks: theirs is relative to a selection that ignores impartiality, yours to the best arguer."

A adds that judge accuracy rises with debater strength (Khan et al., 2024), so a random right-side debater raises ε_step. The program officer pointed you to Procaccia's work and asked how yours differs; a social-choice reader will look here first.

Fix, p. 10 (two sentences; the mechanism stays as it is).

- Replace "With uniform draws, (b) gives 1/|E_y| unless the judge's group holds a*_y; with two or three groups per answer, this beats the 1/(1+m) guaranteed by randomizing over would-be winners when one agent is chosen and each rates m others (Kurokawa et al., 2015)." with "With uniform draws, (b) gives 1/|E_y| unless the judge's group holds a*_y, a guarantee that falls as more groups give the answer; splitting the groups at random into candidates and raters, as in Alon et al. (2011), would guarantee ½ whenever two or more groups give it, and we will compare the two."
- Delete "; a judge lottery that caps any agent's chance at p̄ lets f colluders hold the seat with probability at most f p̄, so judge chances are kept as equal as possible (Flanigan et al., 2021)", so the sentence ends at "…best arguer low." The judge is already drawn uniformly from the agents that did not answer, so the cap adds nothing.
- Why not switch mechanisms this week: the first version of this review proposed the split, and its wording had three errors a social-choice reviewer would catch. Raters must exclude non-answering members of the candidates' groups; both halves must be nonempty (if only the candidate half must be, a*_y can be chosen with probability 4/9 < ½); and one model group must not hold both debater seats. Name the split as the comparison and work it out under Q2.

Cut: the judge-lottery clause pays for the longer first sentence.

C10. "Why trust the judge?" is answered in pieces, and Figure 1 never shows the threat Tasks 1.2 and 2.2 exist for · should-fix

Where: §3.3 Overview, p. 11; Proposition 4, p. 12; Figure 1, p. 2.

Reviewer (A): "Impartiality removes the judge's stake, not its errors. Reliability rests on Proposition 4(iii), which holds roughly for tool-checkable steps and not otherwise, and the proposal estimates the debate's error rate (p. 9) and the judge's error by kind of step (p. 12) without tying either to who may judge. In Figure 1 the compromised agent argues the wrong side, which ordinary debate already handles; a colluder arguing the right side badly, or a colluding judge, never appears."

Fix:

- p. 11. Replace "The judge need not be wiser than the debaters: it rules only on single steps, reproduces tool evidence itself, is drawn impartially (Task 1.2), and is audited (Task 2.2)." with "Impartiality removes the judge's stake, not its errors. The judge need not be wiser than the debaters: it rules only on single steps, and its error on each kind of step is measured on planted debates; it settles tool-checkable steps by re-running them, so its error there approaches the tool's; and a judge whose audited rulings go wrong more often than those rates predict is flagged and barred (Task 2.2)."
- p. 12. Replace the harness paragraph's last sentence, "In the vessel example, A4 disputes A3's step "so V left at 06:00" with a 09:10 port report listing V at its berth, and the judge re-runs A4's query and picks no (Figure 1).", with "By a union bound, $\varepsilon_{step} \leqslant P(\text{not (i)}) + P(\text{not (ii)}) + k\,\varepsilon_J$, where ε_J is the judge's error per ruling and k the number of rulings (two without recursion), so each harness part lowers one term we measure." Re-running lowers ε_J, false-step search lowers P(not (ii)), and the right of reply lowers P(not (i)); Figure 1 already shows the vessel example.
- Figure 1 caption: C52.

Cut: the bound replaces the vessel-example sentence; the longer p. 11 paragraph is paid for in the Space plan.

C11. Recent work a reviewer will name · should-fix

Where: F3, p. 1; §2 debate paragraph, p. 5; Task 2.1 evaluation, p. 12; Task 2.2, p. 13.

Reviewer (A): "A collaborative protocol that locates and examines the point of disagreement beat standard debate for non-expert judges, 62.1% to 49.2% (Jiang et al., 2026); H3 pits debate against discussion, so it is the obvious baseline. The F3 evidence is overstated: in Xu (2026), open-weight judges (7B–32B) shown a success log accepted 78–90% of failed outputs, while frontier judges were essentially unaffected. The inspection-game framing has an AI-control precedent (Griffin et al., 2024)."

Fix: C50 (F3) and C61 (§2). On p. 12, "and replace debate with open discussion" → "and replace debate with open discussion or collaborative disagreement resolution". On p. 13, "Treating auditing as an inspection game," → "Treating auditing as an inspection game, as AI-control games do (Griffin et al., 2024),". Optional: Gardner-Challis et al. (2026) analyze monitors that collude by recognizing their own model's outputs, which supports grouping agents by underlying model in Task 1.2.

Cut: §5, p. 19. Shorten the service sentence to "The PI has served as an area chair for ACL, EMNLP, and COLING (2023–2025) and co-organized the AI4Research workshop at AAAI 2025." The full list is in the CV.

C12. The overlap boundary is vague where reviewers will look hardest · should-fix

Where: §4 "Relation to other support", p. 19; §5, p. 20.

Reviewer (N or program): "The PI holds an Open Philanthropy grant on 'control evaluations of AI debate', and Task 2.2 with its red team is a control evaluation of AI debate. 'This proposal funds aims those awards do not' does not say which aims. A pending NSF proposal also trains reward models on OpenReview reviews and runs paper code, as testbed C does."

Fix (same length). Replace "the Open Philanthropy grant supports control evaluations of AI debate; the Amazon Research Award concerns process-guided fine-tuning. This proposal funds aims those awards do not, and any overlap with a pending proposal would be removed with the program officers." with "the Open Philanthropy grant supports [its scope in one clause] and does not fund [the RG1–RG3 aims it excludes]; the Amazon Research Award concerns process-guided fine-tuning. The pending NSF proposals study explainable single-model reasoning and hypothesis generation; testbed C uses OpenReview papers only as questions for collective evaluation."

Fill the brackets yourself; I can't verify the grant's scope. Your CV lists it as awarded ("Control Evaluations for AI Debate", Oct 2025–present), so "holds" on p. 20 is right. In main-full-reviewed.tex the brackets hold a red stand-in that you must replace.

C13. Success criteria have no numbers, and the Navy payoff has no metric · optional

Where: Table 4, p. 18; §3.5 Metrics, p. 17; Table 2 testbed B, p. 16.

Reviewer (N): "Table 4's criteria ('a margin … that grows', 'below a preset bound') are not checkable, and no metric is stated in the running example's own terms. The maritime scenarios are fully synthetic."

Fix:

- Table 4, G1. Replace "A margin over majority vote that grows with how widely mistakes are shared, where three or more groups err independently" with "At least [5] points over majority vote when 3 of 8 agents share a source, growing with sharing; none with fewer than three independent groups". Pick a number you will defend; Figure 2(a) shows 11.7 points at six copies in simulation.
- Table 4, G3. Append "; forged tool outputs accepted in under [5]% of debates".
- §3.5 Metrics, p. 17. Add: "In testbed B we also report the running example's metric: patrols sent in error per 100 vessel questions, at each audit budget."
- Table 2, testbed B. "including synthetic maritime scenarios" → "including maritime scenarios built on public U.S. Coast Guard AIS tracks (Marine Cadastre) with synthetic reports and forgeries". Update the DMP's data-origin line to match.

Space: C13 uses 2 of the 4 spare lines in the tested plan, so it needs no further cut. If you adopt it, cite the AIS data with the bibliography entry in the Space plan.

3. Math audit

All seven statements are correct under their stated assumptions, and none is new as mathematics. Three carry real weight (Proposition 2, Proposition 6, and eq. (1) once fixed); the rest are design lemmas.

Item	Correct?	New or known (credited?)	Design choice it justifies	M reads it as
Eq. (1)	Yes, given calibration	Standard decomposition; with (3), Chow's rule (no)	Task-to-term map, ablations, τ	Decorative now; load-bearing with ε_cal (C8)
Prop. 1	Yes	Immediate from Jaffe et al.'s model (no)	Groups enter through one shared answer	Decorative, honest
Prop. 2	Yes; two gaps	Corollary of Allman et al. 2009 (yes)	At least 3 independent groups; H1	Load-bearing
Prop. 3	(a) yes; (b) yes but weak	Partition-mechanism argument (partly)	Roles drawn before ratings; outsiders rate	(a) load-bearing; (b) unfavorable

Item	Correct?	New or known (credited?)	Design choice it justifies	M reads it as
Prop. 4	Yes, immediate	One-step soundness of debate (yes)	Losing rule, step disputes, re-running	Decorative; load-bearing with corollary
Prop. 5	Yes	Optional skipping, inspection games (partly)	R4: audits random, never triggered	Trivial but load-bearing for R4
Prop. 6	Yes for binary; text around it no	Cai et al. 2025; Natarajan et al. 2013 (no)	Leave-group-out reference	Load-bearing once corrected

C14. Eq. (1), p. 6

- **Correct?** Yes, as a bound: the law of total probability plus a union bound, with the bracketed terms read as probabilities given a debate. The first term needs calibration on questions with $\pi_q \geqslant \tau$.
- **Counterexample to the framing.** Two colluding groups back the wrong answer on a targeted question and push π_q to 0.97, above $\tau = 0.9$. No debate occurs and the decision is wrong, yet the first term charges at most 0.1 and no other term applies; only the calibration assumption saves the inequality. C8 adds the missing term.
- **Assumptions for LLM agents.** A fitted posterior is miscalibrated exactly where the model is misspecified (an undetected shared factor) or attacked. Recalibration on development questions fixes honest shift only.
- **New?** No. Splitting error into "decided alone" and "deferred" is the standard selective-prediction decomposition, and (3) is Chow's (1970) rule with an imperfect expert (Mozannar and Sontag, 2020). Neither is credited (C4).
- **Justifies:** the map from tasks to error terms, the per-term ablations, and τ.
- **Panel reading:** decorative as written, an accounting identity. With ε_{cal}, and one line showing that (3) minimizes the loss built on (1) (C8), it becomes the proposal's organizing objective.

C15. Proposition 1, p. 8

- **Correct?** Yes. The group's likelihood ratio $[(1-\lambda)r + \lambda]/[\lambda r + (1-\lambda)]$ increases in r when $\lambda < 1/2$ and runs from $\lambda/(1-\lambda)$ to $(1-\lambda)/\lambda$. The $|Y|$-answer bound checks too, for $\lambda \leqslant (|Y| -1)/|Y|$.
- **Assumptions.** One hidden shared answer per question, members independent given it. That fits agents reading one stale source, and fits same-model agents only partly, since their shared errors vary by question. It also assumes the partition is right: a copy group split in two counts twice, and an honest agent merged into a group loses its vote.
- **A subtlety worth one clause.** If λ_g is estimated near 0, the cap is large: a group whose source is usually right counts as one very reliable agent. That is correct, but it is not "about once" in the everyday sense the text uses.
- **New?** It follows at once from Jaffe et al.'s model, as a data-processing bound through one latent variable. Not credited.
- **Justifies:** modeling a group through a shared latent answer, so copies cannot outvote one source.
- **Panel reading:** decorative but honest; a sentence would do.

C16. Proposition 2, p. 8

- **Correct?** Yes, under its assumptions. It is a corollary of Allman, Matias and Rhodes (2009), who group observed variables into three conditionally independent blocks to apply Kruskal's theorem. Three blocks of full column rank meet Kruskal's condition for any number of answers, and the two-group non-identifiability is right too.
- **Two gaps, one clause each.** (1) With answering agents drawn at random, "at least three groups answer each question" is not enough: require that the same three groups co-answer a positive share of questions, or that each group shares questions with two groups already identified. (2) "Up to relabeling": say what fixes the labels; your planted cases with known answers do.
- **Assumptions.** Independence across groups given the true answer is shaky even across model families, since Kim et al. (2025) find correlated errors across providers. It fails for colluders (C8), and the result is asymptotic; rates are Q1.
- **New?** Known, and credited (Kruskal; Allman et al.).
- **Justifies:** the pool design (at least three independent families) and H1's prediction that the gain vanishes with two groups.
- **Panel reading:** load-bearing; the only proposition that makes a falsifiable prediction. Keep it and add the two clauses.

C17. Proposition 3, p. 10

- **(a) Correct?** Yes. The judge, the debated answers and the group draws ignore ratings, and a drawn group is rated only by outsiders. Say once that the property concerns ratings only: an agent can still change its chance of a role through its own answer, as in the literature.
- **(b) Correct?** Yes, but weak: 1/|E_y|, and with one-member groups the representative is uniformly random. The cited 1/(1+m) is right — Credible Subset achieves k/(k+m), so 1/(1+m) at k = 1 (Kurokawa et al., 2015) — but it is measured against a different benchmark.
- **New?** (a) is the partition-mechanism argument of Alon et al. (2011) for a fixed partition, and extending it to collusion within a group is immediate. I found no prior statement of the group version, but it adds little.
- **Justifies:** drawing roles before any rating, and rating only by outsiders.
- **Panel reading:** (a) load-bearing for impartiality; (b) an unfavorable bound on display. A random split of groups into candidates and raters would give ½ (C9).

C18. Proposition 4, p. 12

- **Correct?** Yes, and immediate: (ii) and (iii) together contain the conclusion.
- **Assumptions.** (ii) is the obfuscated-arguments problem: the honest side must find the false step in an argument the other side can make long. Prover–estimator debate addresses it (Brown-Cohen et al., 2025), which you cite only for recursion. (iii) holds roughly for tool-checkable steps, not textual ones, and (i) fails with true but selective facts, as you note.
- **New?** The one-step soundness argument of debate (Irving et al., 2018), credited in the sentence before.
- **Justifies:** the losing rule, step-level disputes, and re-running; each harness part maps to one assumption.
- **Panel reading:** decorative as stated. C10′s bound, ε_step ⩽ P(not (i)) + P(not (ii)) + kε_J for k rulings, makes it load-bearing: three measured terms, one per harness part.

C19. Proposition 5, p. 13

- **Correct?** Yes. Audit draws are independent of the past, so on the colluder′s adaptively chosen decisions they stay independent with probability ρ each (optional skipping), and n escapes have probability (1−ρ)^n.

- **What it does not show.** Being audited is not being flagged. A colluder whose excess error stays inside honest noise is audited and never flagged, so the title "random audits cannot be dodged" overclaims; the load-bearing claim is the harm bound, which is only conjectured.
- **The conjectured rate.** An anytime-valid test must leave room of order √(t log log t) after t audits, by the law of the iterated logarithm (Howard et al., 2021). Excess wrong decisions before flagging should then be of order √(n log log n / ρ), not √n.
- **Blame.** An honest agent often paired with colluders errs more than its own record predicts, so the α guarantee needs partner effects in the prediction (C24).
- **New?** Standard in inspection games (Avenhaus et al., 2002), which you credit.
- **Justifies:** R4, audits drawn at random and never triggered.
- **Panel reading:** trivial but load-bearing for R4. Fold the proof into the statement: "If each real decision is audited independently with probability ρ, drawn after it is made, a colluder's first n steered decisions all escape audit with probability (1−ρ)^n, whatever it infers about planted cases."

C20. Proposition 6, p. 14

- **Correct?** Yes for yes-or-no questions under REINFORCE. It also holds under GRPO with binary rewards, since a wrong reference flips the sign of every normalized advantage.
- **Wrong or unsupported around it** (C6). The chance term is a baseline; ω_q cancels under GRPO's normalization; the multi-answer weight assumes evenly spread wrong answers; and the pilot's "0.74 that Proposition 6 predicts" is not a prediction it makes, since Proposition 6 bounds the expected per-step update, not the share of the final accuracy gain.
- **Weighting direction.** ω_q = 1 − 2η̂_q multiplies by the push, while the unbiased correction divides by it: Cai et al.'s backward correction is (R̃ − ρ0)/(1 − ρ0 − ρ1). Multiplying is a defensible variance choice; say so, and compare with their corrections, of which they report the forward one is more stable under heavy noise.
- **Assumptions.** The reference's error must not depend on the agent's answer. That holds when the partition is right and fails as agents trained on the same decisions grow alike, as you note. η_q < 1/2 fails on shared misconceptions, where η̂_q from the same posterior is underestimated (Proposition 2's undetectable case), so the held-out labeled check is the only guard.
- **New?** No: the symmetric case of Cai et al. (2025), and of Natarajan et al. (2013). Not credited (C4).

- **Justifies:** the leave-group-out reference, which is the real principle in RG3, and the condition under which training on decisions does not reverse.
- **Panel reading:** load-bearing; it is the answer to the colleague's question (c) for first answers. Credit it and correct the text around it.

4. The "ad hoc" test

The formal framework mostly worked: Tasks 1.1, 2.1 and 2.2, and the first-answer half of Task 3, now read as derived from a stated failure mode. Task 1.2's single-group draw and Task 3's message credit still read as plausible choices.

#	Task	White paper (Oct 23)	Full proposal	Verdict	What would make it principled
C21	1.1	Dawid–Skene "with shared error factors"; agents giving the same unusual answers share a factor. A description, no model	Generative model (2), Props. 1–2, expected-loss rule (3)	Derived at the core; partition search and "audit flags as a prior" are heuristics	Start from Jaffe et al.'s spectral partition recovery (their Lemma 4 gives a score gap under stated conditions); state Q1; say how flags enter, e.g. as an adversarial mixture component weighted by the sequential test's evidence
C22	1.2	"Partition idea of impartial peer review"; top-rated member of a randomly drawn group	Mechanism (4), Prop. 3	Impartiality derived; the single-group draw has no objective and a weak price bound	State the objective (best expected representative subject to impartiality); name the random split of groups as the comparison (C9); Q2 for coalitions
C23	2.1	Harness; "walk through their arguments to the first step on which they differ"; judge re-runs tool evidence	Prop. 4, each harness part tied to an assumption	Derived in structure; false-step search, the "short exchange" and the right of reply are heuristics	C10's bound on ε_step, with each heuristic assigned to one term and its measured failure rate

#	Task	White paper (Oct 23)	Full proposal	Verdict	What would make it principled
C24	2.2	Planted debates; random human audits; flag agents worse on audited than planted cases	Prop. 5, anytime-valid test with α control, blame by random pairings, inspection-game target	Derived for detection error; blame and "rulings that favor one debater" unspecified	Blame as an additive model of agent effects on audited wrong decisions, identifiable because pairings are random, so the α guarantee accounts for partners; Q4 with its rate
C25	3	Each message rewarded for moving the group toward the decision; "tool-backed decisions weigh more"	Prop. 6, leave-group-out reference, chance term, credit (6)	First answers derived after C6; message credit still heuristic (telescoping without invariance; Shapley deferred)	Credit only evidence-settled debates, with the corollary in C6; position (6) against ΔBelief-RL (Auzina et al., 2026), which credits each turn by the change in the agent's own belief, where yours uses the judge's

C26. The colleague's questions

Question	Status	Where	Fix
(a) Reliability without ground truth, with correlated errors and dishonest agents	Partly answered	Prop. 2, p. 8, for correlated errors given the partition and three independent groups; dishonest agents only as "audit flags as a prior", p. 9	C8
(b) Why trust the judge	Partly answered	p. 11 ("The judge need not be wiser…"); Prop. 4(iii), p. 12	C10
(c) Wrong group decisions reinforcing bad behavior	First answers answered; debate messages not	Prop. 6 and leave-group-out, pp. 14–15; stop rule for shared misconceptions, p. 16. Credit (6) rewards moving the judge toward a decision the messages caused	C6

Question	Status	Where	Fix
(d) Concrete evaluation metrics	Metrics answered; targets missing	p. 12; §3.5 Metrics, p. 17; Table 4, p. 18	C13

Question (e), the Navy connection, is Section 7 below.

The program officer's six questions all have a home: the problem (p. 1), why it matters (pp. 1–2), today's methods (§2, Table 1), the approach (§3), limitations (p. 18) and impact (pp. 17–18). Two are weak. "Why should it work" leans on the overclaim fixed in C5, and "Limitations" is four lines that omit the two largest: colluders who win without a debate (C8), and that re-running checks a tool's output, not its source (C34).

C27. Dropped or changed from the white paper

- **Dropped:** "Without trusted audits, dishonest agents that imitate honest ones cannot be told apart from them" (white paper, "Why it will succeed"). The Steinhardt point survives, but the reason audits are needed at all is gone; C8's sentence after Proposition 2 restores it.
- **Dropped:** the impact line "These methods would let conferences and funders try impartial, audited peer review by AI." Peer review itself remains, as testbed C and the p. 16 example; C63 restores the impact line.
- **Dropped:** HiPRAG's measured result (unnecessary searches cut to 2.3%), the one concrete number for a prior system this project reuses. Optional to restore in §5.
- **Changed:** the white paper's "tool-backed decisions weigh more" is gone from Task 3's text, though Figure 1 still says "A4's tool-backed step earns the most credit". C6's message-credit sentence makes the text match the figure, with a reason.
- **Changed:** "the two walk through their arguments to the first step on which they differ" became "each names the first step of the other's argument that it disputes" (p. 12), a short exchange, and a ruling on each disputed step. That fits Proposition 4, but the judge now rules at least twice, hence k = 2 in C10's bound.
- **Changed:** Jaffe et al. (2016) went from a stated limitation ("simple patterns") to a citation for "dependence between classifiers", while Task 1.1 now proposes the same model as new (C4).
- **Kept:** one question with separate tasks, the vessel example, the impartial judge, two kinds of audit, three testbeds, the red team and the baselines. Nothing the program officer approved structurally was lost.

6. Pilots

Run the Task 2.1 pilot first and the Task 1.1 pilot second; both are cheap and back the proposal's central claims. Restate the other three as expected outcomes, with no numbers.

#	Pilot (page)	Would the expected result convince?	Run before Oct 30?
C28	1.1: shared hint, 9 agents, 300 GSM8K problems (p. 9)	Partly: a planted hint is the easiest shared factor to find	Yes, second (inference plus EM)
C29	1.2: strategic rater on PeerEval (p. 10)	Only the 0.92; "not at all" is Prop. 3(a) by construction	No
C30	2.1: forged tool outputs, 200 disputes (p. 12)	Yes, with two controls added	Yes, first (calculator plus a synthetic port table)
C31	2.2: discriminator AUC; simulated flagging (p. 14)	AUC yes; the simulation is a power calculation	The AUC only, if time
C32	3: GRPO on GSM8K, 1.5B model (p. 15)	Yes, if the majority baseline is self-referential	Only with a spare GPU day

C28. Task 1.1 pilot, p. 9

Planted hints give identical wrong answers on a known third of problems. A skeptic wants two more numbers from the same runs: the gain when same-family models share errors with no hint, and the two-group case where Proposition 2 predicts failure. Report calibration error too, since eq. (1) depends on it.

If not run, write: "In a pilot, up to four of nine agents from four model families will receive the same flawed hint on 300 GSM8K problems. We expect the shared-factor model to keep its no-hint accuracy while majority vote and the independent-error model fall, and its advantage to vanish when fewer than three independent groups remain (Proposition 2)."

C29. Task 1.2 pilot, p. 10

"Not at all under our mechanism" restates Proposition 3(a); even measured, it is not evidence, and a reviewer will say so. Only the price (0.92 of the best arguer's quality) is informative, and with one-member groups it mostly measures how alike the agents are.

Write: "On PeerEval's 40,156 judgments, we will measure what a rater gains by rating every rival lowest under plain weighted ranking, a gain our mechanism removes by Proposition 3(a), and the price of impartiality: the chosen representative's argument quality relative to the best arguer's."

C30. Task 2.1 pilot, p. 12

This is the pilot that matters most: it tests the central novelty and gives F3 its own evidence. Re-running defeats a forged output by construction, so the informative numbers are the transcript-only acceptance rate and the residual error when the judge re-runs (wrong calls, misread outputs). Add two controls where re-running cannot help, a genuine but stale record and a true but selective fact (Hu et al., 2026), so R3′s limit is measured rather than assumed.

If not run, write: "In a pilot on 200 disputes that a calculator or a port-database query settles, with forged tool outputs on the wrong side, we expect a transcript-only judge to accept many forgeries and a re-running judge's error to approach the tools' own; stale and selectively chosen evidence are controls where re-running should not help."

C31. Task 2.2 pilot, p. 14

The AUC contrast between templated and edited planted cases is a real empirical question and answers Needham et al. (2025) directly; it needs a strong model and a few hundred real and planted pairs. The flagging simulation follows from the test's design, so present it as a design calculation, which you can compute exactly, not as a pilot result.

If not run, write: "We will test whether a strong model can tell planted cases from real ones, comparing templated cases with controlled edits. For the sequential test, a design calculation gives the audited decisions needed to flag a colluder whose audited decisions go wrong three times as often as predicted."

C32. Task 3 pilot, p. 15

Collapse under one reward and steady gains under the other would be the strongest single answer to the colleague's main worry. Fix three things first:

- Delete "close to the $1 - 2\hat{\eta} = 0.74$ that Proposition 6 predicts" whatever you measure (C20).
- Say whether the majority reward includes the trained model's own samples. The collapse in Shafayat et al. (2025) needs that self-reference; with fixed references, majority agreement converges to the references' majority instead.
- In expectation under GRPO, reward (5) reduces to agreement with the left-out reference: ω_q cancels in the normalization, and the chance term adds no gradient. Credit the leave-group-out principle for whatever you find.

It is the most expensive pilot: three GRPO runs plus reference answers on the training set. If not run, write: "In a pilot training a 1.5B-parameter model on GSM8K with GRPO, with three of eight reference agents sharing a flawed hint, we expect rewards for self-consistent majority answers to raise and then collapse accuracy (Shafayat et al., 2025), and rewards against the left-out reference to raise it without collapse while that reference is right more often than wrong."

C33. Order of work, and Figure 2(a)

- If you can run only one, run Task 2.1. Put each measured result in place of its orange sentence, in the same space.
- Figure 2(a) is not marked as pending (its orange curve is only a line color), but I found no simulation code for it in the repo. Confirm it comes from a run you can reproduce, since §3.7 promises re-runnable results.
- Compile the final PDF with \finalbuildtrue, which stops on any remaining \pending.

7. Naval relevance and skim test

A Navy reader will find the example credible and the work clearly basic research. The gaps: re-running checks a tool's output but not its source, and the Navy payoff has no number.

C34. Is the watch-officer example credible? (pp. 1–2, Figure 1)

Mostly yes. Maritime operations centers fuse schedules, port reports, position tracks and imagery, and you already cite spoofed position broadcasts and untracked vessels. A Navy reader will push on three points:

- **Sources, not only outputs.** Re-running a port-database query proves the debater did not fabricate the output, not that the database or the AIS feed is genuine; a forged source reproduces perfectly. Add to Limitations, p. 18: "Re-running verifies reported evidence, not its source."
- **Time.** The officer needs an answer in minutes. Add half a sentence after (3): the cost c includes delay, so urgent questions skip debate unless the stakes are high. In the tested compile this pushed Section 6 four lines past p. 20, so leave it out unless you free more space.
- **Why five agents.** Tie the setup to the agent-networks bullet already on p. 3, so it reads as the near-term case rather than a contrivance.

Cut: the Limitations sentence is in the tested plan and paid for there.

C35. Basic research with broad use, or a narrow demo?

Basic research with broad use. The methods are general; the testbeds span math and code, multi-document questions, and peer review; and the Navy uses span maritime domain awareness, intelligence analysis, test and evaluation, logistics, and cyber. The risk runs the other way: the Navy appears only in one synthetic testbed, which C13's public AIS tracks and patrol-error metric fix without narrowing the science.

C36. Does the title match what the proposal delivers?

Roughly. "Collective reasoning" is broader than the deliverables, and the title omits what is distinctive: agents that share mistakes or collude, and debate before a judge who re-runs evidence. An option: "Trustworthy Decisions by AI Agents That Share Mistakes or Collude: Weighting, Evidence-Checked Debate, and Audits". Keeping the title the program officer saw on the white paper is also defensible, so treat this as optional.

C37. Skim test: title, page 1 and Figure 1 only

The three sentences a busy panelist carries into the discussion:

1. "She wants a group of AI agents to give one trustworthy answer when their errors are correlated and some are compromised, like five tools disagreeing on whether a ship has left port."
2. "Her pipeline counts copied sources once, debates only the unclear questions before a judge who re-runs the evidence, audits a random sample for collusion, and retrains the agents on the group's decisions."

3. "Well framed, with a credible Navy use, but from page 1 I can't tell what is mathematically new beyond assembling these pieces."

The third sentence is what C4, C5 and C57 exist to change.

Where a non-specialist first gets lost: Figure 1, panel 2, at "share an error factor and count about once" and "{A3} drawn, so A3 argues". The reader does not know why debaters are drawn at random or why the compromised agent may argue; C52 answers both inside the figure. On page 1, F3's "tool calls" and "log" are agent jargon that a physicist or engineer on the panel will skim past.

8. Against the funded examples

Your milestones, success criteria and naval relevance are already stronger in form than Joshi's and Chen's. The gaps a panel would notice are direct preliminary evidence, named open problems, and how your foundational work is framed.

#	What the funded proposals do	Where yours falls short	Worth it in 20 pages? How
C38	Joshi builds each task on her own published estimator or bound and shows it; her Task 1 gives the new estimator's error and plots it against correlation	Your published evidence (PRD, FG-PRM, HiPRAG) is adjacent to the methods; the direct evidence is one simulation plus placeholders	Yes. Run two pilots (C33) and drop adjacent numbers that support no task (the C4 and C5 cuts)
C39	Joshi frames each task by the questions it will answer and calls the hard ones open problems	Your hard mathematics is scattered as "we will characterize" and "we will prove such bounds"	Yes. The Q1–Q4 paragraph (C5), in the tested plan
C40	Joshi's approach carries exact error expressions from her own analyses; Zhang's and Yousefi's carry physical models	Your propositions are one-liners	Partly. No new theorems in a day, but each proposition can carry a number the experiments test: Prop. 2's three-group threshold (H1), C10's ε_step terms, Q4's rate. No extra space

#	What the funded proposals do	Where yours falls short	Worth it in 20 pages? How
C41	Yousefi ties the work to named ONR program emphases and an operational forecasting need	Your 2026 S&T Strategy quote beats Joshi's and Chen's generic paragraphs, but no operational quantity appears	Yes. Patrols sent in error per 100 questions (C13) and the program's own areas (C58); C58 is in the tested plan, and C13 costs 2 spare lines
C42	Joshi's PI section names a foundational paper and the follow-up work it spawned	§5 lists your contributions but not the uptake of PRD, the work this proposal extends; your CV says groups at NYU, Anthropic and Cohere built on it	Yes; your CV states it. Add one clause on PRD's uptake to its §5 bullet (p. 20); cut "IQA-EVAL (Li et al., 2024c) evaluates interactive question answering with persona-driven AI agents.", which supports no task. The cut is in the tested plan; the clause costs 1 of the 4 spare lines

On C42: the program officer told you he looks for independent, post-PhD impact, so this is the line he can repeat to the panel. Nothing to copy on schedules: Joshi's year checkmarks and Chen's three dated milestones are coarser than your Table 3.

9. Overlap and compliance

No task duplicates the CAREER award or the two pending NSF proposals; the overlap a reviewer is likeliest to raise is with your Open Philanthropy grant. The only compliance risk in pp. 1–20 is text below 12 point in figures, tables and captions.

#	Proposal	What it covers	Overlap with this proposal	Boundary drawn?
C43	NSF CAREER (funded): consistent event graphs from long documents	Event extraction, question–answer generation, event knowledge graphs	Testbed B reuses MEQA-style event structures to build maritime scenarios	Yes; §4 says so
C44	NSF "Explainable Methods for Natural Language Reasoning" (pending)	Deductive, abductive and inductive reasoning; builds the MEQA dataset; a reward model for explanations	MEQA only; no debate, judges, collusion or multi-agent RL	Not named; add to C12's sentence
C45	NSF "Automating Scientific Knowledge Discovery" (pending)	Hypothesis generation with reward models trained on OpenReview ratings; LLM-as-judge evaluation; code run in a sandbox for verifiable RL rewards	Testbed C also uses OpenReview papers and re-runs released code; Task 3 is RL	Not named; C12's sentence draws it
C46	Open Philanthropy grant (named on pp. 19–20)	"Control evaluations of AI debate"	Task 2.2 and the red team are control evaluations of AI debate	No; see C12; your CV lists it as awarded (Oct 2025–present)

C47. Fonts

- **Body text** is 12.0-pt Nimbus Roman (via mathptmx), a Times clone. Joshi's, Chen's and Zhang's funded proposals used the same family.
- **Below 12 pt:** captions at 10.9 pt, Table 1's headers at 10 pt and figure text at 8 pt, about 15% of the characters on pp. 1–20. They sit on pp. 2 (Figure 1), 6 (Table 1), 8 (Figure 2), 10 (Figure 3), 11 (Figure 4), 13 (Figure 5) and 16–18 (Tables 2–4).

- **The rule:** the NOFO's format list, highlighted by your grants office, says "Times New Roman, 12-point" with no exemption for figures or tables. Its stated automated check covers file type, macros and passwords, so rejection on fonts is unlikely, but a strict reviewer could note it.
- **Do:** ask Taylor Harris today whether ONR applies the rule to figures and captions. If yes, setting captions and Tables 1–4 at 12 pt costs about 20 lines on top of the plan (compiled). Cutting Figure 5, whose numbers are illustrative, and Figure 3, which Figure 1's second panel repeats, pays for it with 7 lines left; text inside the remaining figures would still be below 12 pt.

C48. Other compliance points

- **Page limit.** Sections 1–6 end on p. 20, and with the tested plan they still do, with about 4 lines to spare (Space plan below).
- **Required parts.** Focus area (cover and §1.4), Future Naval Relevance, schedule and milestones, goals and benchmarks, Gold Standard Science, Management, PI Qualifications and all six Responsibility items are present.
- **Gold Standard Science.** The NOFO asks you to "describe your success in implementing" it, but §3.7 describes plans; "as the PI's previous work does" is the only past claim. Name one or two past releases you can point to in that first sentence, for example PeerEval's first release if its judgments are public.
- **NOFO number.** The NOFO's cover-page instruction prints N0001426SF0004, with an extra 0, while its title page says N0001426SF004. Your cover matches the title page; no change.
- **Order.** The NOFO lists Responsibility after the Letters of Support; you place it as Section 6 so pp. 1–20 stay contiguous. That is reasonable; mention it to Taylor if the portal is strict.
- **Timing.** The NOFO recommends submitting two business days before the Oct 30 deadline, which is today.

10. Line edits

Thirteen sentence-level rewrites, in page order, nearly all on pp. 1–5. Eight are in the tested plan (C50, C52, C55, C57, C58, C60, C61, C62). C49, C53, C56 and C63 are optional and cost nothing; C54 costs 2 of the 4 spare lines. C51 and C59 from the first version are withdrawn.

C49 · p. 1, §1.1, the bold central problem

- Original: "…can reach trustworthy decisions and evaluations."
- Revised: "…can reach trustworthy decisions and evaluations, with errors bounded and measured."
- Why: gives "trustworthy" a meaning the panel can check, and points ahead to eq. (1). Optional; no cost in space.

C50 · p. 1, F3

- Original: "judges shown a log claiming that tool calls succeeded accepted most failed outputs (Xu, 2026)."
- Revised: "open-weight judges shown a log claiming the tool calls succeeded accepted 78–90% of failed outputs (Xu, 2026)."
- Why: the paper found frontier judges essentially unaffected; the narrower claim is the one it supports.

C51 · withdrawn

- It changed Figure 1 for the split mechanism, which this version no longer recommends (C9).

C52 · p. 2, Figure 1 caption

- Original: "RG3 trains every agent on the decision."
- Revised: "RG3 trains every agent on the decision. Debaters are drawn at random, so a compromised agent may still argue: A3 argues the wrong side and loses here, and audits limit how often a compromised agent on the right side can argue it badly (Task 2.2)."
- Why: it answers the first question a non-specialist asks of the figure and shows the threat Tasks 1.2 and 2.2 exist for (C10). A representative defends its own first answer, so A3, which said yes, could never argue no; the first version of this edit said it could.

C53 · pp. 2–3, §1.2, RG3

- Original: "When that reference is more often right than wrong, training follows the agents' true accuracy in expectation, without human labels, so it can keep working after the agents outgrow human judges."
- Revised: "When that reference is more often right than wrong, training on it favors the same answers as training on true answers (Proposition 6), without human labels, so it can keep working after the agents outgrow human judges."
- Why: with more than two answers, single updates need not point toward the truth (C6), but the answers rewarded most are still the true ones. Optional; no cost in space.

C54 · p. 3, §1.2, central hypothesis

- Original: "and learns from its decisions through rewards that agreement alone cannot raise."
- Revised: "and learns from its decisions through rewards scored against references that exclude each agent's own group."
- Why: the leave-group-out reference protects the update; the chance term does not (C6). Optional; costs 2 of the 4 spare lines.

C55 · p. 3, §1.3, What is new (i)

- Original: "(i) A rater model for agents that answer and argue, in which a group sharing a mistake counts at most once and the posterior decides which questions need a debate, even unanimous ones; and an impartial way to choose debaters and judges."
- Revised: "(i) An extension of the dependent-classifier model of Jaffe et al. (2016) to agents that answer, argue, and may be adversarial, whose posterior decides which questions need a debate, even unanimous ones; and an impartial way to choose debaters and judges."

C56 · p. 3, §1.3, What is new (iv)

- Original: "(iv) Self-improvement from group decisions, with first-answer rewards that conformity cannot raise and message credit set by a judge that re-runs evidence."
- Revised: "(iv) Self-improvement from group decisions, with first-answer rewards scored against a reference that excludes the agent's group, and message credit only in debates settled by reproduced evidence." Optional; no cost in space.

C57 · p. 3, §1.3, Why it will succeed

- Original: "Each method follows from a principle whose property we prove for agents that share mistakes, argue, and may collude (Sections 3.2–3.4), and most tasks build on our previous work:"
- Revised: "Each method follows from a stated failure mode and a property proved under stated assumptions (Propositions 1–6), with harder guarantees posed as open questions (Section 3.1), and most tasks build on our previous work:"

C58 · p. 3, §1.4, Focus Area

- Original: "…currently addressed using heuristics or non-principled approaches'' (Office of Naval Research, 2026b)."

- Revised: "…non-principled approaches" (Office of Naval Research, 2026b), in two of its areas: multi-modal, multi-scale information integration and decision making under uncertainty."
- Why: both are among the six research areas on the program's page, so the fit is stated in the program officer's own terms.

C59 · withdrawn

- C34's Limitations sentence makes the same point in less space.

C60 · p. 4, §2, and Table 1, p. 6

- Original (§2): "dependence between classifiers (Jaffe et al., 2016)"
- Revised: "groups of dependent classifiers (Jaffe et al., 2016), the model Task 1.1 extends"
- Original (Table 1 caption): "No current approach fully addresses F1, F3, or F4, and none fully addresses more than one column."
- Revised: "No current approach fully addresses F3 or F4, and none addresses F1 and F2 together." In the "Rater models without an answer key" row, cite Dawid and Skene (1979) and Jaffe et al. (2016), with Dong et al. still cited in §2, and set the F1 mark to ✓.

C61 · p. 5, §2, debate paragraph

- Original: "We know of no debate protocol designed for many agents that share mistakes or collude."
- Revised: "Collaborative disagreement resolution beat debate for non-expert judges (Jiang et al., 2026) but also assumes honest agents; we know of no debate protocol designed for many agents that share mistakes or collude."

C62 · p. 5, §2, self-improvement paragraph

- Original: "and rewarding agreement with other models' majority does not correct for chance agreement and relies on model diversity against shared mistakes (Yang et al., 2026)."
- Revised: "and rewarding agreement with other models' majority relies on model diversity against shared mistakes (Yang et al., 2026)."
- Why: under policy gradient a chance correction does not change the update (C6), so this criticism would rebound on your own reward.

C63 · pp. 17–18, §3.5, Results and new knowledge

- Original: "Scientifically, it will show when a debate among AI agents can be trusted and how few audits keep the agents honest;"
- Revised: "Scientifically, it will show when a debate among AI agents can be trusted and how few audits keep the agents honest, which would let conferences and funders try impartial, audited peer review by AI;"
- Why: restores the white paper's peer-review impact line, a use the program officer raised himself. Optional; no cost in space.

Space: the line edits are paid for by the cuts in the Space plan below. Of the four cuts listed here in the first version, only the Rising Star deletion is kept (cut 3); the others freed no lines.

Space plan (compile-tested)

With C4–C12, the line edits marked as in the plan, and the five cuts below applied to a copy of main-full.tex, Sections 1–6 end on p. 20 with about 4 lines to spare and no overfull boxes. The first version's estimates were off by about 20 lines: a sentence that spills onto a new line costs a whole line, while most small cuts free none. The compiled source is main-full-reviewed.tex, and main-full-review.diff shows every change word by word.

Version compiled	Where Section 6 ends
Your current draft	Last line of p. 20
First version of this review, all fixes	21 lines onto p. 21
Must-fix only (C4–C7, C50, C62), with their own cuts	p. 20, 3 lines spare
Must-fix and should-fix (adds C8–C12, C34, C52, C58, C61)	3 lines onto p. 21
Both, plus the five cuts below	p. 20, 4 lines spare

Cuts in the plan (no content is lost):

1. §3.6, p. 18: merge its two paragraphs into "The project runs 36 months from April 1, 2027. Table 3 gives the schedule and milestones M1–M6, and Table 4 each goal's benchmark and success criterion; misses are reported as findings."

2. §3.5, p. 16, first paragraph → "Every question has a known answer for scoring; in experiments, a human audit reveals it for a decision drawn at random, and we also study audits that are sometimes wrong (Table 2)." The cover page and DMP already say all data are public or synthetic.
3. §5, pp. 19–20: delete the Rising Star sentence, the IQA-EVAL sentence, and "This work has appeared at NeurIPS, ICLR, ACL, EMNLP, and TMLR." All three are in the CV.
4. §3.7, p. 18: run the Gold Standard Science intro and its five tenets as two paragraphs, each tenet keeping its bold run-in head.
5. §6, p. 20: run the six Responsibility items as two paragraphs, each keeping its bold run-in head.

Optional edits, each compiled on top of the plan:

Edit	Spare lines it uses (of 4)
C49, C53, C56, C63	0 each
C42, PRD uptake	1
C13, numbers, metric and AIS data	2
C54	2
C8′s threshold sentence	2
C34′s delay sentence	Does not fit: 4 lines onto p. 21

C13, C42, C53 and C56 together leave about 1 spare line. Recompile after your own changes: with measured pilot results in your original sentences, the plan leaves 2 spare lines instead of 4, and your Open Philanthropy wording can move another. If you need more room, deleting Figure 5 (illustrative numbers) frees 9 lines and Figure 3 (which Figure 1′s second panel repeats) frees 6; the AAAI-26 clause on p. 1 frees none.

New bibliography entries. They cost no page space. The first five are already in main-full-reviewed.tex; the last is needed only with C13, where Table 2 would say "public AIS tracks \citep{uscg2024ais}".

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\bibitem[{Cai et~al.(2025)}]{cai2025noisy} X.-Q. Cai, W. Wang, F. Liu, T.
Liu, G. Niu, and M. Sugiyama (2025). \href{https://arxiv.org/abs/2510.00915}
{Reinforcement Learning with Verifiable yet Noisy Rewards under Imperfect
Verifiers}. arXiv:2510.00915.
\bibitem[{Chow(1970)}]{chow1970optimum} C. K. Chow (1970).
\href{https://doi.org/10.1109/TIT.1970.1054406}{On Optimum Recognition Error
and Reject Tradeoff}. IEEE Transactions on Information Theory 16(1):41--46.
\bibitem[{Griffin et~al.(2024)}]{griffin2024games} C. Griffin, L. Thomson, B.
Shlegeris, and A. Abate (2024). \href{https://arxiv.org/abs/2409.07985}{Games
for AI Control: Models of Safety Evaluations of AI Deployment Protocols}.
arXiv:2409.07985.
\bibitem[{Jiang et~al.(2026)}]{jiang2026disagreement} Y. Jiang, C. Chen, T.
Wu, L. Sun, H. Liu, S. Feng, and C. Tan (2026).
\href{https://arxiv.org/abs/2607.01251}{Collaborative Disagreement Resolution
for Scalable Oversight}. arXiv:2607.01251.
\bibitem[{Mozannar and Sontag(2020)}]{mozannar2020consistent} H. Mozannar and
D. Sontag (2020). \href{https://proceedings.mlr.press/v119/mozannar20b.html}
{Consistent Estimators for Learning to Defer to an Expert}. ICML, PMLR
119:7076--7087.
\bibitem[{U.S. Coast Guard(2024)}]{uscg2024ais} U.S. Coast Guard Navigation
Center (2024). \href{https://www.fisheries.noaa.gov/inport/item/73064}
{Nationwide Automatic Identification System 2024}. Distributed by
MarineCadastre.gov, NOAA Office for Coastal Management.
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Sources

I opened each of these, except where noted, as of the date above. Kohli (2026), Shi et al. (2026) and Shu (2026), cited on pp. 1 and 11, say what the proposal says; Xu (2026) needs C50. Not verified by me: the Open Philanthropy grant's scope (your CV lists it as awarded) and the PRD uptake named in your CV.

- Jaffe et al. (2016), Unsupervised Ensemble Learning with Dependent Classifiers — model checked in the PDF
- Cai et al. (2025), RL with Verifiable yet Noisy Rewards under Imperfect Verifiers — noise identity and both corrections checked
- Natarajan et al. (2013), Learning with Noisy Labels
- Chow (1970), On Optimum Recognition Error and Reject Tradeoff — volume 16, p. 41 confirmed through a citing paper; IEEE Xplore would not open
- Mozannar and Sontag (2020), Consistent Estimators for Learning to Defer to an Expert

- Kurokawa et al. (2015), Impartial Peer Review — k/(k+m) checked in the full text
- Jiang et al. (2026), Collaborative Disagreement Resolution for Scalable Oversight
- Xu (2026), Don't Read the Log
- Griffin et al. (2024), Games for AI Control
- Gardner-Challis et al. (2026), When Can We Trust Untrusted Monitoring?
- Howard et al. (2021), Time-uniform, Nonparametric, Nonasymptotic Confidence Sequences
- Auzina et al. (2026), Intrinsic Credit Assignment for Long Horizon Interaction
- Kohli (2026), Nine Judges, Two Effective Votes
- Shi et al. (2026), Emergent Collusion in Long-Horizon LLM Agent Interaction
- Shu (2026), Unanimity Without Persuasion
- Nationwide Automatic Identification System 2024 (NOAA InPort) — U.S. Coast Guard AIS, public download
- ONR Mathematical Data Science program — the quoted phrase and the six research areas checked
- ONR 2026 S&T Strategy — both phrases quoted on p. 3 checked

From your bibliography, not reopened: Allman et al. (2009), Kruskal (1977), Alon et al. (2011), Avenhaus et al. (2002), Steinhardt et al. (2016), Brown-Cohen et al. (2025), Irving et al. (2018), Khan et al. (2024), Kim et al. (2025), Shafayat et al. (2025), Hu et al. (2026a), Needham et al. (2025), Dawid and Skene (1979), Dong et al. (2009), Whitehill et al. (2009), Flanigan et al. (2021), Yang et al. (2026), Biswas et al. (2026), Zheng et al. (2023), Verga et al. (2024), Du et al. (2024), PRD (Li et al., 2024a), FG-PRM (Li et al., 2025), HiPRAG (Wu et al., 2026), IQA-EVAL (Li et al., 2024c).